Fiscal Year:	FY 2009	**Task Last Updated:**	FY 06/07/2010
PI Name:	Bedford, Joel Ph.D.		
Project Title:	Dose-Rate Effects and Components of Systems Governing Variations in Susceptibility for Carcinogenic and Acute Radiation Risks following Gamma-Ray, Proton, or HZE Irradiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jbedford@colostate.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	970-491-7492
Organization Name:	Colorado State University		
PI Address 1:	Environmental and Radiological Health Sciences		
PI Address 2:	1618 Campus Delivery		
PI Web Page:			
City:	Fort Collins	**State:**	CO
Zip Code:	80523	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	05/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 5/31/2012 (previously 11/30/2011), per NSSC information (Ed., 11/29/2011) NOTE: End date is now 11/30/2011, per NSSC information (Ed., 5/31/2011) NOTE: Received NCE through 5/31/2011, per C. Guidry/JSC (08/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nagasawa , Hatsumi (Colorado State University) Peng, Yuanlin (Colorado State University) Weil, Michael (Colorado State University)		
Grant/Contract No.:	NNX07AP85G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The central hypothesis is that dose-rate effects in the 1 to 0.005 Gy/min range will be the same for protons vs. gamma-rays for chromosomal aberration induction. Second, levels of chromosomal radiosensitivity, and their dose-rate dependence will be similar for cells from individuals with genetic differences in radiosensitivity. Such a biomarker may help identify differences among individuals in a component contributing to their susceptibility to these effects. The aims for the year reported below corresponding to FY2009 were: 1) to compare dose-rate effects for gamma-rays vs. protons for chromosomal aberration induction in cultured human fibroblast and epithelial cells. (This would test assumptions regarding equivalence of gamma-ray and high energy protons with respect to RBEs and dose-rate effects of interest for astronauts.) 2) to determine whether changes in the 2D culture or 3D tissue-like context of cells may affect the utility of the putative markers for predicting carcinogenic and acute effects radiosensitivity. For low passage normal human fibroblasts and for MCF10A human mammary epithelial cells, the dose rate effect for induction of chromosomal aberrations seen for gamma-rays in the range of 70cGy/min (HDR) to 0.005cGy/min (LDR) was virtually identical for protons with energies of 1000MeV or 150MeV. For the MCF10A cells the responses (aberration induction, cell killing and gamma H2AX formation at low dose rates) in 2D or 3D culture were identical.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of variation among individuals in their susceptibility to radiation carcinogenesis and the genetic determination of the extent of existing variation would be important information, not only of interest to individual astronauts who will be exposed to radiation during space missions, but would also be of interest to anyone who may be exposed to radiation in their occupation or in connection with activities or experiences where radiation exposure may be involved.
Task Progress:	Dose-Rate Effects Addressing the first major aim required preliminary studies to determine the extent of dose rate effects for chromosomal aberration induction in the cell system we are using and most importantly, to determine whether this dose rate effect can be resolved for gamma-rays for dose-rates that could reasonably be expected to be provided for protons at either NSRL at Brookhaven National Laboratory. In our last report, covering the period 9/1/07 to 6/26/08, we presented results of preliminary experiments testing whether a dose rate effect of sufficient magnitude could be demonstrated using gamma-rays comparing irradiations at 2.5 Gy/minute vs. 0.008 Gy/minute (0.48 Gy/h or 48 cGy/hour) using 4 and 8 hour exposure times for the low dose rate and delivering 1.9 and 3.8 Gy respectively at the two dose rates. We did observe a significant dose rate effect for both the low passage human fibroblasts irradiated in the G0/G1 state and plateau phase CHO cells. Based on this result, we participated in two NSRL runs at BNL using protons of two different energies, 1 GeV and 0.15 GeV (150 MeV) with the “low dose rates” slightly lower at 30 cGy per hour for the protons. And we also ran parallel gamma-ray experiments. Our first run was April 19 and 20 2009, and the second run was May 18 and 19, 2009. We used the same dose-rates in both runs and used 2 human cell strains in the first and 4 cell strains in the second. All the chromosomal aberration scoring were not completed during this report period (FY2009) but results from the first run with one cell strain are reported belos. The total doses delivered were 1.0, 2.0, and 3.0 Gy at both high and low dose rates. Aberrations scored included interstitial deletions, terminal deletions, and dicentrics (a dicentric with one accompanying acentric fragment = one dicentric) There were no differences in the dose-rate effects relative to gamma rays for protons of these energies and the RBEs were not significantly different from 1.0 for either the high or the low dose-rates for this cell strain. The dose-response relationships were linear, with a slope of 0.160 aberrations per cell per Gy for the HDR and a slope of 0.077 aberrations per cell per Gy for the LDR exposures. The slopes of the dose response curves were not significantly different for gamma-rays and the protons either for the HDR or LDR exposures. Radiation Responses of Cells in 2D Monolayer vs. 3D Ductal Structures We have developed the MCF10A mammary epithelial cell system used extensively by Bissell and colleagues, and Barcellos-Hoff and colleagues to facilitate study of cellular responses in a culture system that more closely simulates the “tissue context” of cells. These cells can be grown in monolayer 2D culture, much like any other cell in the traditional or classical cell culture situation, or the same cells can be grown in matrigel with certain growth factors, such that the cells form ductal acinar structures that closely simulate milk-producing mammary ducts. Since it has been strongly asserted that cells in such 3D tissue like contexts “respond very differently than in a 2D monolayer context” we have done preliminary experiments to compare the formation and disappearance of gamma-H2AX after irradiation for the same cells in the 2D vs 3D condition. Our results have now confirmed that there is no difference for the same cells in the two conditions measuring several different end-points of cell responses in the 2D and 3D systems. Damage end-points measured included cell killing, chromosomal aberration induction, gamma-H2AX formation after acute dose irradiation, and after continuous low dose rate irradiation, with care being taken to rule out complicating factors that might be result from cell cycle and PLD repair effects.
Bibliography Type:	Description: (Last Updated: 02/04/2013)
Articles in Peer-reviewed Journals	Lin YF, Nagasawa H, Peng Y, Chuang EY, Bedford JS. "Comparison of several radiation effects in human MCF10A mammary epithelial cells cultured as 2D monolayers or 3D acinar structures in matrigel." Radiat Res. 2009 Jun;171(6):708-15. PMID: 19580477 , Jun-2009